

ELLIPSOMETRY

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16.1 GLOSSARY

A	instrument matrix
D_ϕ	film thickness period
d	film thickness
E	electrical field
$\mathbf{E}_0$	constant complex vector
$f()$	function
I	interface scattering matrix
k	extinction coefficient
L	layer scattering matrix
N	complex refractive index = $n - jk$
n	real part of the complex refractive index
R	reflection coefficient
r	reflection coefficient
S_{ij}	scattering matrix elements
s, p	subscripts for polarization components
X	$\exp(-j2\pi d/D_\phi)$
Δ	ellipsometric angle
ϵ	dielectric function
$\langle \epsilon \rangle$	psuedo dielectric function
ρ	$R_p/R_s = \tan \psi \exp(j\Delta) = \chi_i/\chi_r$
ϕ	angle of incidence
χ_i	E_{is}/E_{ip}
χ_r	E_{rs}/E_{rp}
ψ	ellipsometric angle

16.2 INTRODUCTION

Ellipsometry is a nonperturbing optical technique that uses the change in the state of polarization of light upon reflection for the in-situ and real-time characterization of surfaces, interfaces, and thin films. In this chapter we provide a brief account of this subject with an emphasis on modeling and instrumentation. For extensive coverage, including applications, the reader is referred to several monographs,^{1–4} handbook,⁵ collected reprints,⁶ conference proceedings,^{7–15} and general and topical reviews.^{16–32}

In ellipsometry, a collimated beam of monochromatic or quasi-monochromatic light, which is polarized in a known state, is incident on a sample surface under examination, and the state of polarization of the reflected light is analyzed. From the incident and reflected states of polarization, ratios of complex reflection coefficients of the surface for the incident orthogonal linear polarizations parallel and perpendicular to the plane of incidence are determined. These ratios are subsequently related to the structural and optical properties of the ambient-sample interface region by invoking an appropriate model and the electromagnetic theory of reflection. Finally, model parameters of interest are determined by solving the resulting inverse problem.

In ellipsometry, one of the two copropagating orthogonally polarized waves can be considered to act as a reference for the other. Inasmuch as the state of polarization of light is determined by the superposition of the orthogonal components of the electric field vector, an ellipsometer may be thought of as a common-path polarization interferometer. And because ellipsometry involves only relative amplitude and relative phase measurements, it is highly accurate. Furthermore, its sensitivity to minute changes in the interface region, such as the formation of a submonolayer of atoms or molecules, has qualified ellipsometry for many applications in surface science and thin-film technologies.

In a typical scheme, Fig. 1, the incident light is linearly polarized at a known but arbitrary azimuth and the reflected light is elliptically polarized. Measurement of the ellipse of polarization of the reflected light accounts for the name ellipsometry, which was first coined by Rothen.³³ (For a discussion of light polarization, the reader is referred to Chap. 12 in this volume. For a historical background on ellipsometry, see Rothen³⁴ and Hall.³⁵)

For optically isotropic structures, ellipsometry is carried out only at oblique incidence. In this case, if the incident light is linearly polarized with the electric vector vibrating parallel p or perpendicular s to the plane of incidence, the reflected light is likewise p - and s -polarized, respectively. In other words, the p and s linear polarizations are the eigenpolarizations of reflection.³⁶ The associated eigenvalues are the complex amplitude reflection coefficients R_p and R_s . For an arbitrary input state with phasor electric-field components E_{ip} and E_{is} , the corresponding field components of the reflected light are given by

$$E_{rp} = R_p E_{ip} \quad E_{rs} = R_s E_{is} \quad (1)$$

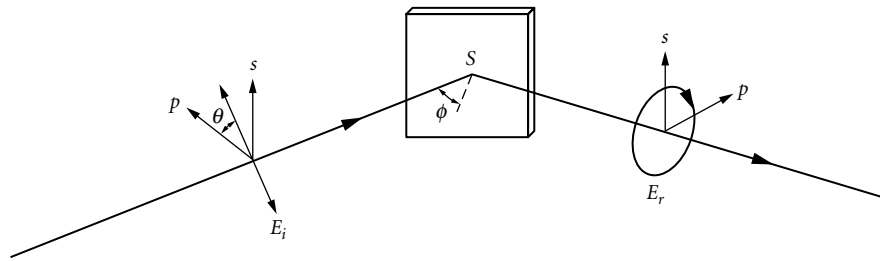


FIGURE 1 Incident linearly polarized light of arbitrary azimuth θ is reflected from the surface S as elliptically polarized. p and s identify the linear polarization directions parallel and perpendicular to the plane of incidence and form a right-handed system with the direction of propagation. ϕ is the angle of incidence.

By taking the ratio of the respective sides of these two equations, one gets

$$\rho = \chi_i / \chi_r \quad (2)$$

where

$$\rho = R_p / R_s \quad (3)$$

$$\chi_i = E_{is} / E_{ip} \quad \chi_r = E_{rs} / E_{rp} \quad (4)$$

χ_i and χ_r of Eqs. (4) are complex numbers that succinctly describe the incident and reflected polarization states of light;³⁷ their ratio, according to Eqs. (2) and (3), determines the ratio of the complex reflection coefficients for the p and s polarizations. Therefore, ellipsometry involves pure polarization measurements (without account for absolute light intensity or absolute phase) to determine ρ . It has become customary in ellipsometry to express ρ in polar form in terms of two *ellipsometric angles* ψ and Δ ($0 \leq \psi \leq 90^\circ$, $0 \leq \Delta < 360^\circ$) as follows

$$\rho = \tan \psi \exp(j\Delta) \quad (5)$$

$\tan \psi = |R_p| / |R_s|$ represents the relative amplitude attenuation and $\Delta = \arg(R_p) - \arg(R_s)$ is the differential phase shift of the p and s linearly polarized components upon reflection.

Regardless of the nature of the sample, ρ is a function,

$$\rho = f(\phi, \lambda) \quad (6)$$

of the angle of incidence ϕ and the wavelength of light λ . Multiple-angle-of-incidence ellipsometry^{38–43} (MAIE) involves measurement of ρ as a function of ϕ , and spectroscopic ellipsometry^{3,22,27–31} (SE) refers to the measurement of ρ as a function of λ . In variable-angle spectroscopic ellipsometry⁴³ (VASE) the ellipsometric function ρ of the two real variables ϕ and λ is recorded.

16.3 CONVENTIONS

The widely accepted conventions in ellipsometry are those adopted at the 1968 Symposium on Recent Developments in Ellipsometry following discussions of a paper by Muller.⁴⁴ Briefly, the electric field of a monochromatic plane wave traveling in the direction of the z axis is taken as

$$\mathbf{E} = \mathbf{E}_0 \exp(-j2\pi N z / \lambda) \exp(j\omega t) \quad (7)$$

where $\mathbf{E}_0$ is a constant complex vector that represents the transverse electric field in the $z = 0$ plane, N is the complex refractive index of the optically isotropic medium of propagation, ω is the angular frequency, and t is the time. N is written in terms of its real and imaginary parts as

$$N = n - jk \quad (8)$$

where $n > 0$ is the refractive index and $k \geq 0$ is the extinction coefficient. The positive directions of p and s before and after reflection form a right-handed coordinate system with the directions of propagation of the incident and reflected waves, Fig. 1. At normal incidence ($\phi = 0$), the p directions in the incident and reflected waves are antiparallel, whereas the s directions are parallel. Some of the consequences of these conventions are as follows:

1. At normal incidence, $R_p = -R_s$, $\rho = -1$, and $\Delta = \pi$.
2. At grazing incidence, $R_p = R_s$, $\rho = 1$, and $\Delta = 0$.
3. For an abrupt interface between two homogeneous and isotropic semi-infinite media, Δ is in the range $0 \leq \Delta \leq \pi$, and $0 \leq \psi \leq 45^\circ$.

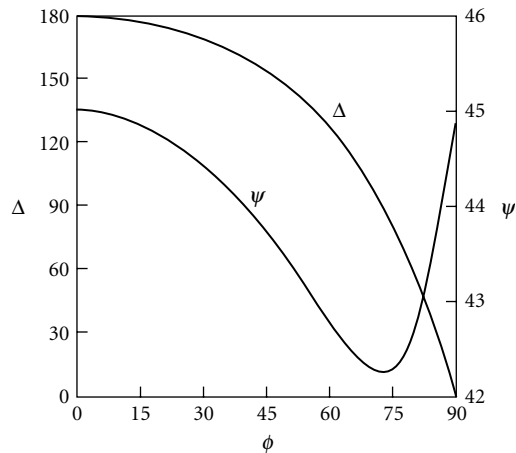


FIGURE 2 Ellipsometric parameters ψ and Δ of an air/Au interface as functions of the angle of incidence ϕ . The complex refractive index of Au is assumed to be $0.306 - j2.880$ at 564-nm wavelength. ψ , Δ , and ϕ are in degrees.

As an example, Fig. 2 shows ψ and Δ vs. ϕ for light reflection at the air/Au interface, assuming $N = 0.306 - j2.880$ for Au⁴⁵ at $\lambda = 564$ nm.

16.4 MODELING AND INVERSION

The following simplifying assumptions are usually made or implied in conventional ellipsometry: (1) the incident beam is approximated by a monochromatic plane wave; (2) the ambient or incidence medium is transparent and optically isotropic; (3) the sample surface is a plane boundary; (4) the sample (and ambient) optical properties are uniform laterally but may change in the direction of the normal to the ambient-sample interface; (5) the coherence length of the incident light is much greater than its penetration depth into the sample; and (6) the light-sample interaction is linear (elastic), hence frequency-conserving.

Determination of the ratio of complex reflection coefficients is rarely an end in itself. Usually, one is interested in more fundamental information about the sample than is conveyed by ρ . In particular, ellipsometry is used to characterize the optical and structural properties of the interfacial region. This requires that a stratified-medium model (SMM) for the sample under measurement be postulated that contains the sample physical parameters of interest. For example, for visible light, a polished Si surface in air may be modeled as an optically opaque (semi-infinite) Si substrate which is covered with a SiO_2 film, with the Si and SiO_2 phases assumed uniform, and the air/ SiO_2 and SiO_2 /Si interfaces considered as parallel planes. This is often referred to as the three-phase model. More complexity (and more layers) can be built into this basic SMM to represent such finer details as the interfacial roughness and phase mixing, a damage surface layer on Si caused by polishing, or the possible presence of an outermost contamination film. Effective medium theories^{46–54} (EMTs) are used to calculate the dielectric functions of mixed phases based on their microstructure and component volume fractions; and the established theory of light reflection by stratified structures^{55–60} is employed to calculate the ellipsometric function for an assumed set of model parameters. Finally, values of the model parameters are sought that best match the measured and computed values of ρ . Extensive data (obtained, e.g., using VASE) is required to determine the parameters of more complicated samples. The latter task, called the

inverse problem, usually employs linear regression analysis,^{61–63} which yields information on parameter correlations and confidence limits. Therefore, the full practice of ellipsometry involves, in general, the execution and integration of three tasks: (1) polarization measurements that yield ratios of complex reflection coefficients, (2) sample modeling and the application of electromagnetic theory to calculate the ellipsometric function, and (3) solving the inverse problem to determine model parameters that best match the experimental and theoretically calculated values of the ellipsometric function.

Confidence in the model is established by showing that complete spectra can be described in terms of a few wavelength-independent parameters, or by checking the predictive power of the model in determining the optical properties of the sample under new experimental conditions.²⁷

The Two-Phase Model

For a single interface between two homogeneous and isotropic media, 0 and 1, the reflection coefficients are given by the Fresnel formulas¹

$$r_{01p} = (\epsilon_1 S_0 - \epsilon_0 S_1) / (\epsilon_1 S_0 + \epsilon_0 S_1) \quad (9)$$

$$r_{01s} = (S_0 - S_1) / (S_0 + S_1) \quad (10)$$

in which

$$\epsilon_i = N_i^2 \quad i = 0, 1 \quad (11)$$

is the dielectric function (or dielectric constant at a given wavelength) of the i th medium,

$$S_i = (\epsilon_i - \epsilon_0 \sin^2 \phi)^{1/2} \quad (12)$$

and ϕ is the angle of incidence in medium 0 (measured from the interface normal). The ratio of complex reflection coefficients which is measured by ellipsometry is

$$\rho = [\sin \phi \tan \phi - (\epsilon - \sin^2 \phi)^{1/2}] / [\sin \phi \tan \phi + (\epsilon - \sin^2 \phi)^{1/2}] \quad (13)$$

where $\epsilon = \epsilon_1 / \epsilon_0$. Solving Eq. (13) for ϵ gives

$$\epsilon_1 = \epsilon_0 \{ \sin^2 \phi + \sin^2 \phi \tan^2 \phi [(1 - \rho) / (1 + \rho)]^2 \} \quad (14)$$

For light incident from a medium (e.g., vacuum, air, or an inert ambient) of known ϵ_0 , Eq. (14) determines, concisely and directly, the complex dielectric function ϵ_1 of the reflecting second medium in terms of the measured ρ and the angle of incidence ϕ . This accounts for an important application of ellipsometry as a means of determining the optical properties (or *optical constants*) of bulk absorbing materials and opaque films. This approach assumes the absence of a transition layer or a surface film at the two-media interface. If such a film exists, ultrathin as it may be, ϵ_1 as determined by Eq. (14) is called the pseudo dielectric function and is usually written as $\langle \epsilon_1 \rangle$. Figure 3 shows lines of constant ψ and lines of constant Δ in the complex ϵ plane at $\phi = 75^\circ$.

The Three-Phase Model

This often-used model, Fig. 4, consists of a single layer, medium 1, of parallel-plane boundaries which is surrounded by two similar or dissimilar semi-infinite media 0 and 2. The complex amplitude reflection coefficients are given by the Airy-Drude formula^{64,65}

$$R = (r_{01v} + r_{12v} X) / (1 + r_{01v} r_{12v} X) \quad v = p, s \quad (15)$$

$$X = \exp[-j2\pi(d/D_\phi)] \quad (16)$$

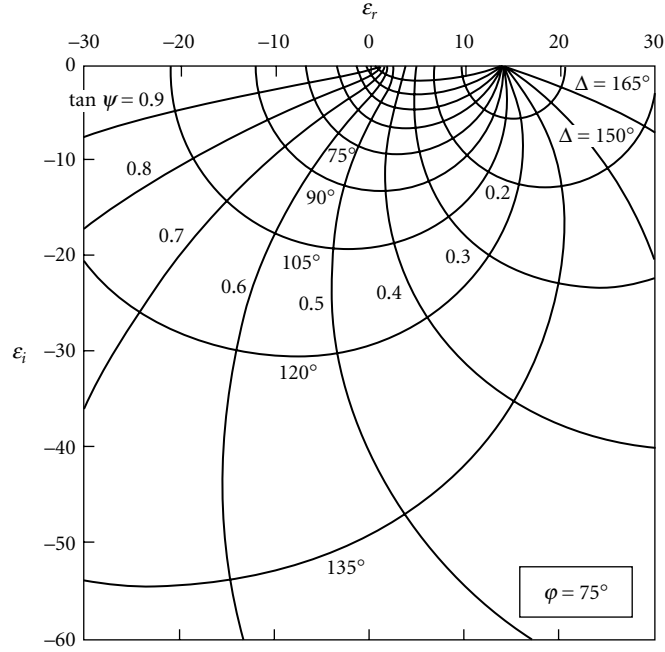


FIGURE 3 Contours of constant $\tan \psi$ and constant Δ in the complex plane of the relative dielectric function ϵ of a transparent medium/absorbing medium interface.

r_{ijv} is the Fresnel reflection coefficient of the ij interface ($ij = 01$ and 12) for the v polarization, d is the layer thickness, and

$$D_\phi = (\lambda/2)(1/S_1) \quad (17)$$

where λ is the vacuum wavelength of light and S_1 is given by Eq. (12). The ellipsometric function of this system is

$$\rho = (A + BX + CX^2)/(D + EX + FX^2) \quad (18)$$

$$\begin{aligned} A &= r_{01p} & B &= r_{12p} + r_{01p}r_{01s}r_{12s} & C &= r_{12p}r_{01s}r_{12s} \\ D &= r_{01s} & E &= r_{12s} + r_{01p}r_{01s}r_{12p} & F &= r_{12s}r_{01p}r_{12p} \end{aligned} \quad (19)$$

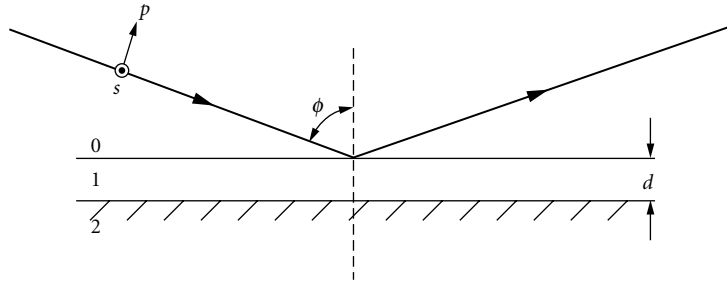


FIGURE 4 Three-phase, ambient-film-substrate system.

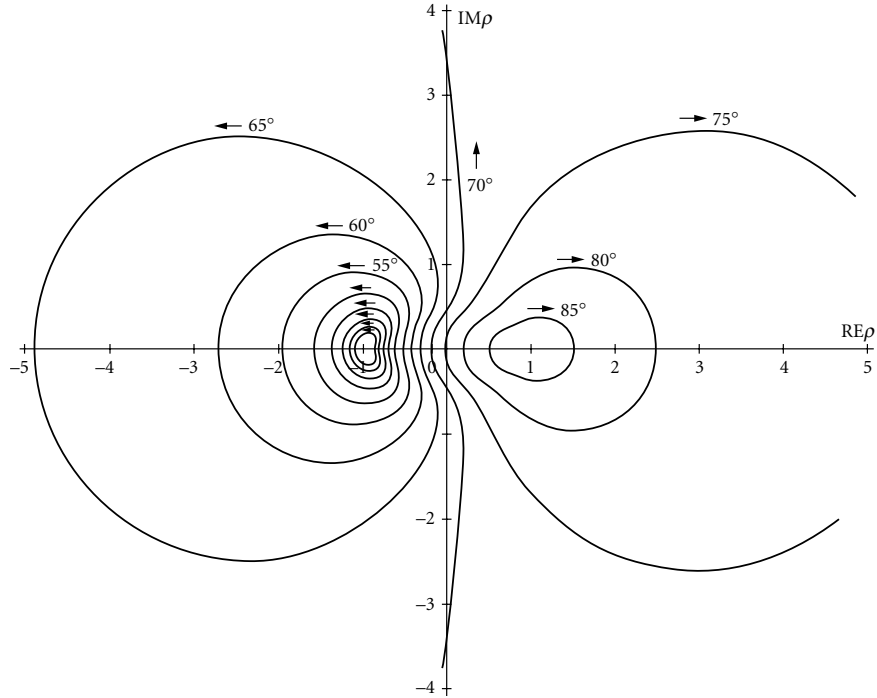


FIGURE 5 Family of constant-angle-of-incidence contours of the ellipsometric function ρ in the complex plane for light reflection in air by the SiO_2/Si film-substrate system at 633-nm wavelength. The contours are for angles of incidence from 30° to 85° in steps of 5° . The arrows indicate the direction of increasing film thickness.⁶⁶

For a transparent film, and with light incident at an angle ϕ such that $\epsilon_1 > \epsilon_0 \sin^2 \phi$ so that total reflection does not occur at the 01 interface, D_ϕ is real, and X , R_p , R_s , and ρ become periodic functions of the film thickness d with period D_ϕ . The locus of X is the unit circle in the complex plane and its multiple images through the conformal mapping of Eq. (18) at different values of ϕ give the constant-angle-of-incidence contours of ρ . Figure 5 shows a family of such contours⁶⁶ for light reflection in air by the SiO_2 -Si system at 633-nm wavelength at angles from 30° to 85° in steps of 5° . Each and every value of ρ , corresponding to all points in the complex plane, can be realized by selecting the appropriate angle of incidence and the SiO_2 film thickness (within a period).

If the dielectric functions of the surrounding media are known, the dielectric function ϵ_1 and thickness d of the film are obtained readily by solving Eq. (18) for X ,

$$X = \{-(B - \rho E) \pm [(B - \rho E)^2 - 4(C - \rho F)(A - \rho D)]^{1/2}\} / 2(C - \rho F) \quad (20)$$

and requiring that^{66,67}

$$|X| = 1 \quad (21)$$

Equation (21) is solved for ϵ_1 as its only unknown by numerical iteration. Subsequently, d is given by

$$d = [-\arg(X) / 2\pi] D_\phi + m D_\phi \quad (22)$$

where m is an integer. The uncertainty of an integral multiple of the film thickness period is often resolved by performing measurements at more than one wavelength or angle of incidence and requiring that d be independent of λ or ϕ .

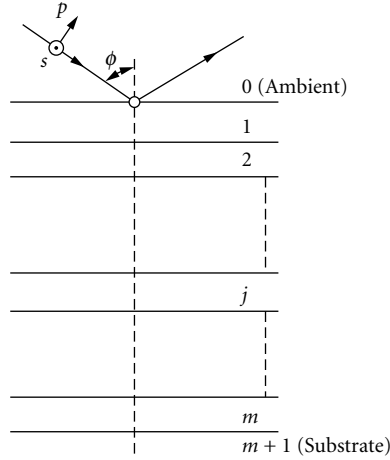


FIGURE 6 Light reflection by a multilayer structure.¹

When the film is absorbing (semitransparent), or the optical properties of one of the surrounding media are unknown, more general inversion methods⁶⁸⁻⁷² are required which are directed toward minimizing an error function of the form

$$f = \sum_{i=1}^N [(\psi_{im} - \psi_{ic})^2 + (\Delta_{im} - \Delta_{ic})^2] \quad (23)$$

where ψ_{im} , ψ_{ic} and Δ_{im} , Δ_{ic} denote the i th measured and calculated values of the ellipsometric angles, and N is the total number of independent measurements.

Multilayer and Graded-Index Films

For an outline of the matrix theory of multilayer systems refer to Chap. 7, "Optical Properties of Films and Coatings," in Vol. IV. For our purposes, we consider a multilayer structure, Fig. 6, that consists of m plane-parallel layers sandwiched between semi-infinite ambient and substrate media (0 and $m + 1$, respectively). The relationships between the field amplitudes of the incident (i), reflected (r), and transmitted (t) plane waves for the p or s polarizations are determined by the scattering matrix equation⁷³

$$\begin{bmatrix} E_i \\ E_r \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \begin{bmatrix} E_t \\ 0 \end{bmatrix} \quad (24)$$

The complex-amplitude reflection and transmission coefficients of the entire structure are given by

$$\begin{aligned} R &= E_r/E_i = S_{21}/S_{11} \\ T &= E_t/E_i = 1/S_{11} \end{aligned} \quad (25)$$

The scattering matrix $\mathbf{S} = (S_{ij})$ is obtained as an ordered product of all the interface $\mathbf{I}$ and layer $\mathbf{L}$ matrices of the stratified structure,

$$\mathbf{S} = \mathbf{I}_{01} \mathbf{L}_1 \mathbf{I}_{12} \mathbf{L}_2 \cdots \mathbf{I}_{(j-1)j} \mathbf{L}_j \cdots \mathbf{L}_m \mathbf{I}_{m(m+1)} \quad (26)$$

and the numbering starts from layer 1 (in contact with the ambient) to layer m (adjacent to the substrate) as shown in Fig. 6. The interface scattering matrix is of the form

$$\mathbf{I}_{ab} = (1/t_{ab}) \begin{bmatrix} 1 & r_{ab} \\ r_{ab} & 1 \end{bmatrix} \quad (27)$$

where r_{ab} is the local Fresnel reflection coefficient of the $ab[j(j+1)]$ interface evaluated [using Eqs. (9) and (10) with the appropriate change of subscripts] at an incidence angle in medium a which is related to the external incidence angle ϕ in medium 0 by Snell's law. The associated interface transmission coefficients for the p and s polarizations are

$$\begin{aligned} t_{abp} &= 2(\epsilon_a \epsilon_b)^{1/2} S_a / (\epsilon_b S_a + \epsilon_a S_b) \\ t_{abs} &= 2S_a / (S_a + S_b) \end{aligned} \quad (28)$$

where S_j is defined in Eq. (12). The scattering matrix of the j th layer is

$$\mathbf{L}_j = \begin{bmatrix} Y_j^{-1} & 0 \\ 0 & Y_j \end{bmatrix} \quad (29)$$

$$Y_j = X_j^{1/2} \quad (30)$$

and X_j is given by Eqs. (16) and (17) with the substitution $d = d_j$ for the thickness, and $\epsilon_1 = \epsilon_j$ for the dielectric function of the j th layer.

Except in Eqs. (28), a polarization subscript $v = p$ or s has been dropped for simplicity. In reflection and transmission ellipsometry, the ratios $\rho_r = R_p/R_s$ and $\rho_t = T_p/T_s$ are measured. Inversion for the dielectric functions and thicknesses of some or all of the layers requires extensive data, as may be obtained by VASE, and linear regression analysis to minimize the error function of Eq. (23).

Light reflection and transmission by a graded-index (GRIN) film is handled using the scattering matrix approach described here by dividing the inhomogeneous layer into an adequately large number of sublayers, each of which is approximately homogeneous. In fact, this is the most general approach for a problem of this kind because analytical closed-form solutions are only possible for a few simple refractive-index profiles.⁷⁴⁻⁷⁶

Dielectric Function of a Mixed Phase

For a microscopically inhomogeneous thin film that is a mixture of two materials, as may be produced by coevaporation or cosputtering, or a thin film of one material that may be porous with a significant void fraction (of air), the dielectric function is determined using EMTs.⁴⁶⁻⁵⁴ When the scale of the inhomogeneity is small relative to the wavelength of light, and the domains (or grains) of different dielectrics are of nearly spherical shape, the dielectric function of the mixed phase ϵ is given by

$$\frac{\epsilon - \epsilon_h}{\epsilon + 2\epsilon_h} = v_a \frac{\epsilon_a - \epsilon_h}{\epsilon_a + 2\epsilon_h} + v_b \frac{\epsilon_b - \epsilon_h}{\epsilon_b + 2\epsilon_h} \quad (31)$$

where ϵ_a and ϵ_b are the dielectric functions of the two component phases a and b with volume fractions v_a and v_b and ϵ_h is the host dielectric function. Different EMTs assign different values to ϵ_h . In the Maxwell Garnett EMT,^{47,48} one of the phases, say b , is dominant ($v_b \gg v_a$) and $\epsilon_h = \epsilon_b$. This reduces the second term on the right-hand side of Eq. (31) to zero. In the Bruggeman EMT,⁴⁹ v_a and v_b are comparable, and $\epsilon_h = \epsilon$, which reduces the left-hand side of Eq. (31) to zero.

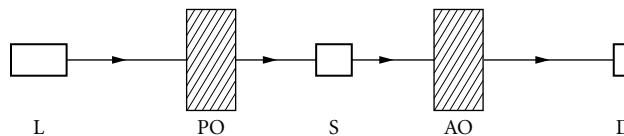


FIGURE 7 Generic ellipsometer with polarizing optics PO and analyzing optics AO. L and D are the light source and photodetector, respectively.

16.5 TRANSMISSION ELLIPSOMETRY

Although ellipsometry is typically carried out on the reflected wave, it is possible to also monitor the state of polarization of the transmitted wave, when such a wave is available for measurement.^{77–81} For example, by combining reflection and transmission ellipsometry, the thickness and complex dielectric function of an absorbing film between transparent media of the same refractive index (e.g., a solid substrate on one side and an index-matching liquid on the other) can be obtained analytically.^{79,80} Polarized light transmission by a multilayer was discussed previously under “Multilayer and Graded-Index Films.” Transmission ellipsometry can be carried out at normal incidence on optically anisotropic samples to determine such properties as the natural or induced linear, circular, or elliptical birefringence and dichroism. However, this falls outside the scope of this chapter.

16.6 INSTRUMENTATION

Figure 7 is a schematic diagram of a generic ellipsometer. It consists of a source of collimated and monochromatic light L, polarizing optics PO on one side of the sample S, and polarization analyzing optics AO and a (linear) photodetector D on the other side. An apt terminology²⁵ refers to the PO as a polarization state generator (PSG) and the AO plus D as a polarization state detector (PSD).

Figure 8 shows the commonly used polarizer-compensator-sample-analyzer (PCSA) ellipsometer arrangement. The PSG consists of a linear polarizer with transmission-axis azimuth P and a linear retarder, or compensator, with fast-axis azimuth C . The PSD consists of a single linear polarizer, that functions as an analyzer, with transmission-axis azimuth A followed by a photodetector D.

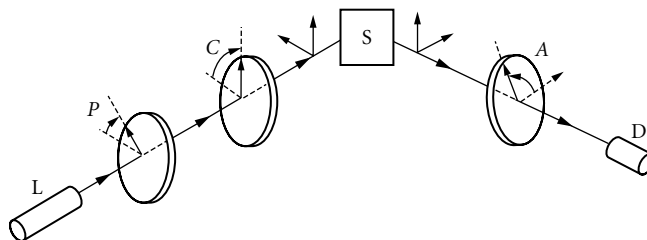


FIGURE 8 Polarizer-compensator-sample-analyzer (PCSA) ellipsometer. The azimuth angles P of the polarizer, C of the compensator (or quarter-wave retarder), and A of the analyzer are measured from the plane of incidence, positive in a counterclockwise sense when looking toward the source.¹

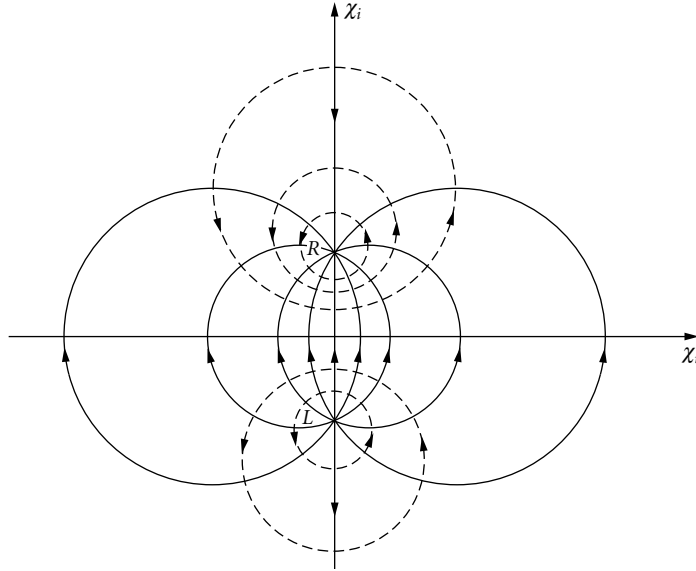


FIGURE 9 Constant C , variable P contours (continuous lines), and constant $P - C$, variable C contours (dashed lines) in the complex plane of polarization for light transmitted by a polarizer-compensator (PC) polarization state generator.¹

All azimuths P , C , and A , are measured from the plane of incidence, positive in a counterclockwise sense when looking toward the source. The state of polarization of the light transmitted by the PSG and incident on S is given by

$$\chi_i = [\tan C + \rho_c \tan(P - C)] / [1 - \rho_c \tan C \tan(P - C)] \quad (32)$$

where $\rho_c = T_{cs}/T_{cf}$ is the ratio of complex amplitude transmittances of the compensator for incident linear polarizations along the slow s and fast f axes. Ideally, the compensator functions as a quarter-wave retarder (QWR) and $\rho_c = -j$. In this case, Eq. (32) describes an elliptical polarization state with major-axis azimuth C and ellipticity angle $-(P - C)$. (The tangent of the ellipticity angle equals the minor-axis-to-major-axis ratio and its sign gives the handedness of the polarization state, positive for right-handed states.) All possible states of total polarization χ_i can be generated by controlling P and C . Figure 9 shows a family of constant C , variable P contours (continuous lines) and constant $P - C$, variable C contours (dashed lines) as orthogonal families of circles in the complex plane of polarization. Figure 10 shows the corresponding contours of constant P and variable C . The points R and L on the imaginary axis at $(0, +1)$ and $(0, -1)$ represent the right- and left-handed circular polarization states, respectively.

Null Ellipsometry

The PCSA ellipsometer of Fig. 8 can be operated in two different modes. In the null mode, the output signal of the photodetector D is reduced to zero (a minimum) by adjusting the azimuth angles P of the polarizer and A of the analyzer with the compensator set at a fixed azimuth C . The choice $C = \pm 45^\circ$ results in rapid convergence to the null. Two independent nulls are reached for each compensator setting. The two nulls obtained with $C = +45^\circ$ are usually referred to as the nulls in zones 2 and 4; those for $C = -45^\circ$ define zones 1 and 3. At null, the reflected polarization is linear and is

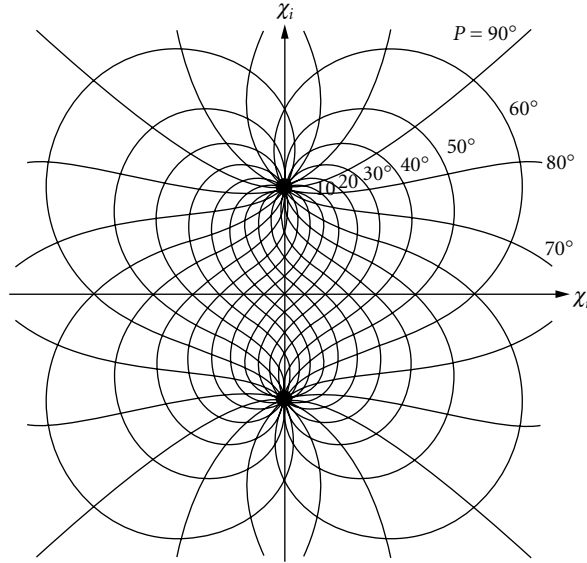


FIGURE 10 Constant P , variable C contours in the complex plane of polarization for light transmitted by a polarizer-compensator (PC) polarization state generator.¹

crossed with the transmission axis of the analyzer; therefore, the reflected state of polarization is given by

$$\chi_r = -\cot A \quad (33)$$

where A is the analyzer azimuth at null. With the incident and reflected polarizations determined by Eqs. (32) and (33), the ratio of complex reflection coefficients of the sample for the p and s linear polarizations ρ is obtained by Eq. (2). Whereas a single null is sufficient to determine ρ in an ideal ellipsometer, results from multiple nulls (in two or four zones) are usually averaged to eliminate the effect of small component imperfections and azimuth-angle errors. Two-zone measurements are also used to determine ρ of the sample and ρ_c of the compensator simultaneously.^{82–84} The effects of component imperfections have been considered extensively.⁸⁵

The null ellipsometer can be automated by using stepping or servo motors^{86,87} to rotate the polarizer and analyzer under closed-loop feedback control; the procedure is akin to that of nulling an ac bridge circuit. Alternatively, Faraday cells can be inserted after the polarizer and before the analyzer to produce magneto-optical rotations in lieu of the mechanical rotation of the elements.^{88–90} This reduces the measurement time of a null ellipsometer from minutes to milliseconds. Large ($\pm 90^\circ$) Faraday rotations would be required for limitless compensation. Small ac modulation is often added for the precise localization of the null.

Photometric Ellipsometry

The polarization state of the reflected light can also be detected photometrically by rotating the analyzer^{91–95} of the PCSA ellipsometer and performing a Fourier analysis of the output signal I of the linear photodetector D. The detected signal waveform is simply given by

$$I = I_0(1 + \alpha \cos 2A + \beta \sin 2A) \quad (34)$$

and the reflected state of polarization is determined from the normalized Fourier coefficients α and β by

$$\chi_r = [\beta \pm (1 - \alpha^2 - \beta^2)^{1/2}] / (1 + \alpha) \quad (35)$$

The sign ambiguity in Eq. (35) indicates that the rotating-analyzer ellipsometer (RAE) cannot determine the handedness of the reflected polarization state. In the RAE, the compensator is not essential and can be removed from the input PO (i.e., the PSA instead of the PCSA optical train is used). Without the compensator, the incident linear polarization is described by

$$\chi_i = \tan P \quad (36)$$

Again, the ratio of complex reflection coefficients of the sample ρ is determined by substituting Eqs. (35) and (36) in Eq. (2). The absence of the wavelength-dependent compensator makes the RAE particularly qualified for SE. The dual of the RAE is the rotating-polarizer ellipsometer which is suited for real-time SE using a spectrograph and a photodiode array that are placed after the fixed analyzer.³¹

A photometric ellipsometer with no moving parts, for fast measurements on the microsecond time scale, employs a photoelastic modulator^{96–100} (PEM) in place of the compensator of Fig. 8. The PEM functions as an oscillating-phase linear retarder in which the relative phase retardation is modulated sinusoidally at a high frequency (typically 50 to 100 kHz) by establishing an elastic ultrasonic standing wave in a transparent solid. The output signal of the photodetector is represented by an infinite Fourier series with coefficients determined by Bessel functions of the first kind and argument equal to the retardation amplitude. However, only the dc, first, and second harmonics of the modulation frequency are usually detected (using lock-in amplifiers) and provide sufficient information to retrieve the ellipsometric parameters of the sample.

Numerous other ellipsometers have been introduced²⁵ that employ more elaborate PSDs. For example Fig. 11 shows a family of rotating-element photopolarimeters²⁵ (REP) that includes the RAE. The column on the right represents the Stokes vector and the fat dots identify the Stokes

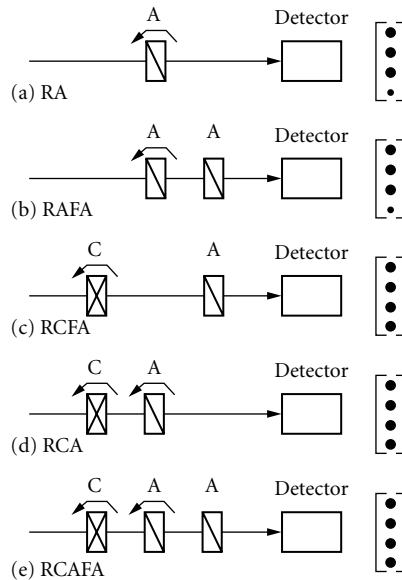


FIGURE 11 Family of rotating-element photopolarimeters (REP) and the Stokes parameters that they can determine.²⁵

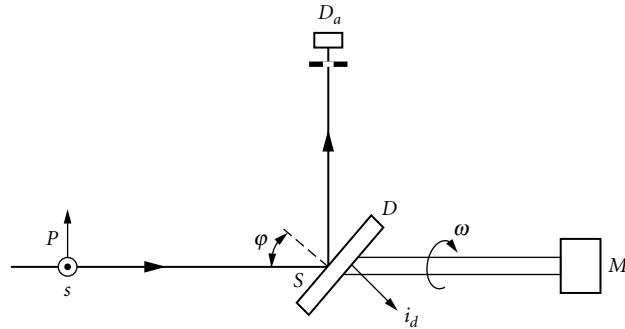


FIGURE 12 Rotating-detector ellipsometer (RODE).¹⁰²

parameters that are measured. (For a discussion of the Stokes parameters, see Chap. 12 in this volume of the *Handbook*.) The simplest complete REP, that can determine all four Stokes parameters of light, is the rotating-compensator fixed-analyzer (RCFA) photopolarimeter originally invented to measure skylight polarization.¹⁰¹ The simplest handedness-blind REP for totally polarized light is the rotating-detector ellipsometer^{102,103} (RODE), Fig. 12, in which the tilted and partially reflective front surface of a solid-state (e.g., Si) detector performs as polarization analyzer.

Ellipsometry Using Four-Detector Photopolarimeters

A new class of fast PSDs that measure the general state of partial or total polarization of a quasi-monochromatic light beam is based on the use of four photodetectors. Such PSDs employ the division of wavefront, the division of amplitude, or a hybrid of the two, and do not require any moving parts or modulators. Figure 13 shows a division-of-wavefront photopolarimeter (DOWP)¹⁰⁴ for

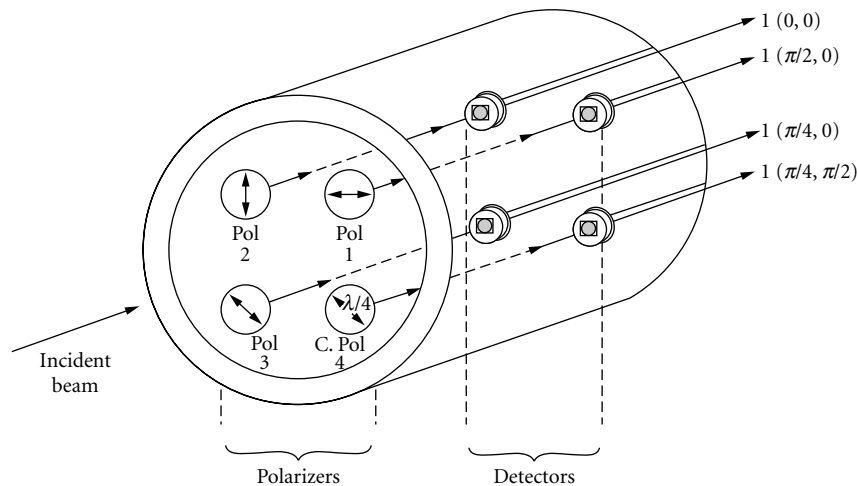


FIGURE 13 Division-of-wavefront photopolarimeter for the simultaneous measurement of all four Stokes parameters of light.¹⁰⁴

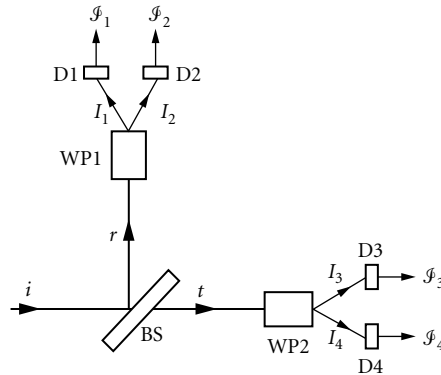


FIGURE 14 Division-of-amplitude photopolarimeter (DOAP) for the simultaneous measurement of all four Stokes parameters of light.¹⁰⁷

performing ellipsometry with nanosecond laser pulses. The DOWP has been adopted recently in commercial automatic polarimeters for the fiber-optics market.^{105,106}

Figure 14 shows a division-of-amplitude photopolarimeter^{107,108} (DOAP) with a coated beam splitter BS and two Wollaston prisms WP1 and WP2, and Fig. 15 represents a recent implementation¹⁰⁹ of that technique. The multiple-beam-splitting and polarization-altering properties of grating diffraction are also well-suited for the DOAP.^{110,111}

The simplest DOAP consists of a spatial arrangement of four solid-state photodetectors Fig. 16, and no other optical elements. The first three detectors (D_0 , D_1 , and D_2) are partially specularly reflecting and the fourth (D_3) is antireflection-coated. The incident light beam is steered in such a way that the plane of incidence is rotated between successive oblique-incidence reflections, hence

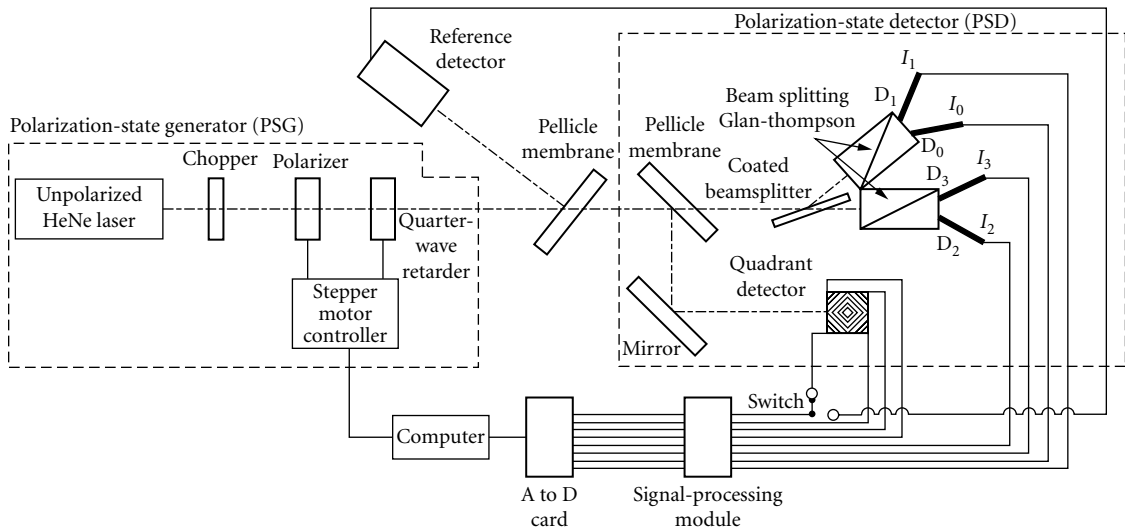


FIGURE 15 Recent implementation of DOAP.¹⁰⁹

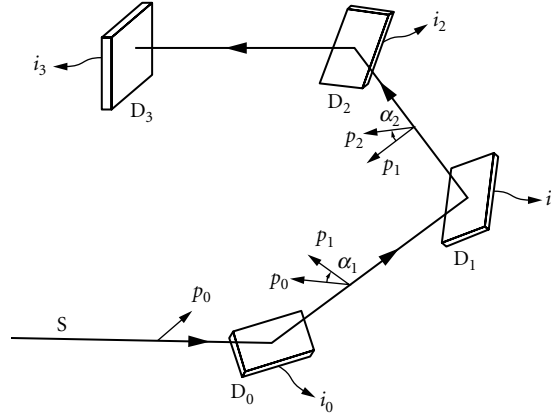


FIGURE 16 Four-detector photopolarimeter for the simultaneous measurement of all four Stokes parameters of light.¹¹²

the light path is nonplanar. In this four-detector photopolarimeter^{112–117} (FDP), and in other DOAPs, the four output signals of the four linear photodetectors define a current vector $\mathbf{I} = [I_0 \ I_1 \ I_2 \ I_3]^t$ which is linearly related,

$$\mathbf{I} = \mathbf{A}\mathbf{S} \quad (37)$$

to the Stokes vector $\mathbf{S} = [S_0 \ S_1 \ S_2 \ S_3]^t$ of the incident light, where t indicates the matrix transpose. The 4×4 instrument matrix $\mathbf{A}$ is determined by calibration¹¹⁵ (using a PSG that consists of a linear polarizer and a quarter-wave retarder). Once $\mathbf{A}$ is determined, $\mathbf{S}$ is obtained from the output signal vector by

$$\mathbf{S} = \mathbf{A}^{-1}\mathbf{I} \quad (38)$$

where $\mathbf{A}^{-1}$ is the inverse of $\mathbf{A}$. When the light under measurement is totally polarized (i.e., $S_0^2 = S_1^2 + S_2^2 + S_3^2$), the associated complex polarization number is determined in terms of the Stokes parameters as¹¹⁸

$$\chi = (S_2 + jS_3)/(S_0 + S_1) = (S_0 - S_1)/(S_2 - jS_3) \quad (39)$$

For further information on polarimetry, see Chap. 15 in this volume.

Ellipsometry Based on Azimuth Measurements Alone

Measurements of the azimuths of the elliptic vibrations of the light reflected from an optically isotropic surface, for two known vibration directions of incident linearly polarized light, enable the ellipsometric parameters of the surface to be determined at any angle of incidence. If θ_i and θ_r represent the azimuths of the incident linear and reflected elliptical polarizations, respectively, then^{119–121}

$$\tan 2\theta_r = (2 \tan \theta_i \tan \psi \cos \Delta) / (\tan^2 \psi - \tan^2 \theta_i) \quad (40)$$

A pair of measurements $(\theta_{i1}, \theta_{r1})$ and $(\theta_{i2}, \theta_{r2})$ determines ψ and Δ via Eq. (40). The azimuth of the reflected polarization is measured precisely by an ac-null method using an ac-excited Faraday

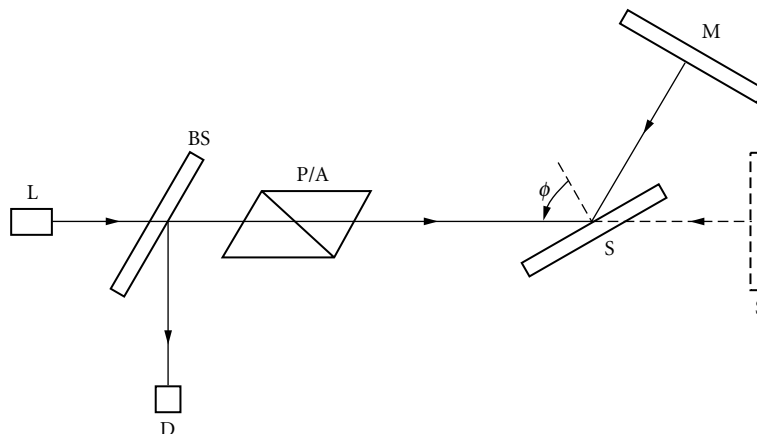


FIGURE 17 Return-path ellipsometer. The dashed lines indicate the configuration for perpendicular-incidence ellipsometry on optically anisotropic samples.¹²⁶

cell followed by a linear analyzer.¹¹⁹ The analyzer is rotationally adjusted to zero the fundamental-frequency component of the detected signal; this aligns the analyzer transmission axis with the minor or major axis of the reflected polarization ellipse.

Return-Path Ellipsometry

In a return-path ellipsometer (RPE), Fig. 17, an optically isotropic mirror *M* is placed in, and perpendicular to, the reflected beam. This reverses the direction of the beam, so that it retraces its path toward the source with a second reflection at the test surface *S* and second passage through the polarizing/analyzing optics *P/A*. A beam splitter *BS* sends a sample of the returned beam to the photodetector *D*. The RPE can be operated in the null or photometric mode.

In the simplest RPE,^{122,123} the *P/A* optics consists of a single linear polarizer whose azimuth and the angle of incidence are adjusted for a zero detected signal. At null, the angle of incidence is the principal angle, hence $\Delta = \pm 90^\circ$, and the polarizer azimuth equals the principal azimuth, so that the incident linearly polarized light is reflected circularly polarized. Null can also be obtained at a general and fixed angle of incidence by adding a compensator to the *P/A* optics. Adjustment of the polarizer azimuth and the compensator azimuth or retardance produces the null.^{124,125} In the photometric mode,¹²⁶ an element of the *P/A* is modulated periodically and the detected signal is Fourier-analyzed to extract ψ and Δ .

RPEs have the following advantages: (1) the same optical elements are used as polarizing and analyzing optics; (2) only one optical port or window is used for light entry into and exit from the chamber in which the sample may be mounted; and (3) the sensitivity to surface changes is increased because of the double reflection at the sample surface.

Perpendicular-Incidence Ellipsometry

Normal-incidence reflection from an optically isotropic surface is accompanied by a trivial change of polarization due to the reversal of the direction of propagation of the beam (e.g., right-handed circularly polarized light is reflected as left-handed circularly polarized). Because this change of polarization is not specific to the surface, it cannot be used to determine the properties of the

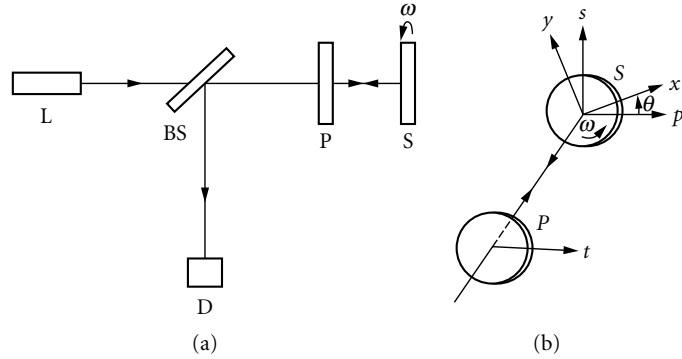


FIGURE 18 Normal-incidence rotating-sample ellipsometer (NIRSE).¹²⁸

reflecting structure. This is why ellipsometry of isotropic surfaces is performed at oblique incidence. However, if the surface is optically anisotropic, perpendicular-incidence ellipsometry (PIE) is possible and offers two significant advantages: (1) simpler single-axis instrumentation of the return-path type with common polarizing/analyzing optics, and (2) simpler inversion for the sample optical properties, because the equations that govern the reflection of light at normal incidence are much simpler than those at oblique incidence.^{127,128}

Like RPE, PIE can be performed using null or photometric techniques.^{126–132} For example, Fig. 18 shows a simple normal-incidence rotating-sample ellipsometer¹²⁸ (NIRSE) that is used to measure the ratio of the complex principal reflection coefficients of an optically anisotropic surface S with principal axes x and y . (The incident linear polarizations along these axes are the eigenpolarizations of reflection.) If we define

$$\eta = (R_{xx} - R_{yy}) / (R_{xx} + R_{yy}) \quad (41)$$

then

$$\eta = \{a_2 \pm j[8a_4(1-a_4) - a_2^2]^{1/2}\} / 2(1-a_4) \quad (42)$$

R_{xx} and R_{yy} are the complex-amplitude principal reflection coefficients of the surface, and a_2 and a_4 are the amplitudes of the second and fourth harmonic components of the detected signal normalized with respect to the dc component. From Eq. (41), we obtain

$$\rho = R_{yy} / R_{xx} = (1 - \eta) / (1 + \eta) \quad (43)$$

PIE can be used to determine the optical properties of bare and coated uniaxial and biaxial crystal surfaces.^{127–130,133}

Interferometric Ellipsometry

Ellipsometry using interferometer arrangements with polarizing optical elements has been suggested and demonstrated.^{134–136} Compensators are not required because the relative phase shift is obtained by the unbalance between the two interferometer arms; this offers a distinct advantage for SE. Direct display of the polarization ellipse is possible.^{134–136}

16.7 JONES-MATRIX GENERALIZED ELLIPSOMETRY

For light reflection at an anisotropic surface, the p and s linear polarizations are not, in general, the eigenpolarizations of reflection. Consequently, the reflection of light is no longer described by Eqs. (1). Instead, the Jones (electric) vectors of the reflected and incident waves are related by

$$\begin{bmatrix} E_{rp} \\ E_{rs} \end{bmatrix} = \begin{bmatrix} R_{pp} & R_{ps} \\ R_{sp} & R_{ss} \end{bmatrix} \begin{bmatrix} E_{ip} \\ E_{is} \end{bmatrix} \quad (44)$$

or, more compactly,

$$\mathbf{E}_r = \mathbf{R}\mathbf{E}_i \quad (45)$$

where $\mathbf{R}$ is the nondiagonal reflection Jones matrix. The states of polarization of the incident and reflected waves, described by the complex variables χ_i and χ_r of Eqs. (4), are interrelated by the bilinear transformation^{85,137}

$$\chi_r = (R_{ss}\chi_i + R_{sp}) / (R_{ps}\chi_i + R_{pp}) \quad (46)$$

In generalized ellipsometry (GE), the incident wave is polarized in at least three different states (χ_{i1} , χ_{i2} , χ_{i3}) and the corresponding states of polarization of the reflected light (χ_{r1} , χ_{r2} , χ_{r3}) are measured. Equation (46) then yields three equations that are solved for the normalized Jones matrix elements, or reflection coefficients ratios,¹³⁸

$$\begin{aligned} R_{pp}/R_{ss} &= (\chi_{i2} - \chi_{i1}H) / (-\chi_{r1} + \chi_{r2}H) \\ R_{ps}/R_{ss} &= (H - 1) / (-\chi_{r1} + \chi_{r2}H) \\ R_{sp}/R_{ss} &= (\chi_{i2}\chi_{r1} - \chi_{i1}\chi_{r2}H) / (-\chi_{r1} + \chi_{r2}H) \\ H &= (\chi_{r3} - \chi_{r1})(\chi_{i3} - \chi_{i2}) / (\chi_{i3} - \chi_{i1})(\chi_{r3} - \chi_{r2}) \end{aligned} \quad (47)$$

Therefore, the nondiagonal Jones matrix of any optically anisotropic surface is determined, up to a complex constant multiplier, from the mapping of three incident polarizations into the corresponding three reflected polarizations. A PCSA null ellipsometer can be used. The incident polarization χ_i is given by Eq. (32) and the reflected polarization χ_r is given by Eq. (33). Alternatively, the Stokes parameters of the reflected light can be measured using the RCFA photopolarimeter, the DOAP, or the FDP, and χ_r is obtained from Eq. (39). More than three measurements can be taken to overdetermine the normalized Jones matrix elements and reduce the effect of component imperfections and measurement errors. GE can be performed based on azimuth measurements alone.¹³⁹ The main application of GE has been the determination of the optical properties of crystalline materials.^{138–143}

16.8 MUELLER-MATRIX GENERALIZED ELLIPSOMETRY

The most general representation of the transformation of the state of polarization of light upon reflection or scattering by an object or sample is described by¹

$$\mathbf{S}' = \mathbf{M}\mathbf{S} \quad (48)$$

where $\mathbf{S}$ and $\mathbf{S}'$ are the Stokes vectors of the incident and scattered radiation, respectively, and $\mathbf{M}$ is the real 4×4 Mueller matrix that succinctly characterizes the linear (or elastic) light-sample interaction.

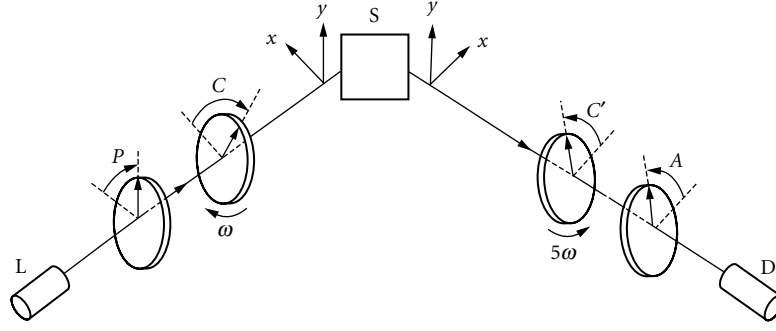


FIGURE 19 Dual-rotating-retarder Mueller-matrix photopolarimeter.¹⁴⁵

For light reflection at an optically isotropic and specular (smooth) surface, the Mueller matrix assumes the simple form¹⁴⁴

$$\mathbf{M} = r \begin{bmatrix} 1 & a & 0 & 0 \\ a & 1 & 0 & 0 \\ 0 & 0 & b & c \\ 0 & 0 & -c & b \end{bmatrix} \quad (49)$$

In Eq. (49), r is the surface power reflectance for incident unpolarized or circularly polarized light, and a , b , c are determined by the ellipsometric parameters ψ and Δ as:

$$a = -\cos 2\psi \quad b = \sin 2\psi \cos \Delta \quad \text{and} \quad c = \sin 2\psi \sin \Delta \quad (50)$$

and satisfy the identity $a^2 + b^2 + c^2 = 1$.

In general (i.e., for an optically anisotropic and rough surface), all 16 elements of $\mathbf{M}$ are nonzero and independent.

Several methods for Mueller matrix measurements have been developed.^{25,145–149} An efficient scheme^{145–147} uses the PCSC'A ellipsometer with symmetrical polarizing (PC) and analyzing (C'A) optics, Fig. 19. All 16 elements of the Mueller matrix are encoded onto a single periodic detected signal by rotating the quarter-wave retarders (or compensators) C and C' at angular speeds in the ratio 1:5. The output signal waveform is described by the Fourier series

$$I = a_0 + \sum_{n=1}^{12} (a_n \cos nC + b_n \sin nC) \quad (51)$$

where C is the fast-axis azimuth of the slower of the two retarders, measured from the plane of incidence. Table 1 gives the relations between the signal Fourier amplitudes and the elements of the

TABLE 1 Relations Between Signal Fourier Amplitudes and Elements of the Scaled Mueller Matrix $\mathbf{M}'$

n	0	1	2	3	4	5	6
a_n	$m'_{11} + \frac{1}{2}m'_{12}$ $+ \frac{1}{2}m'_{21} + \frac{1}{4}m'_{22}$	0	$\frac{1}{2}m'_{12} + \frac{1}{4}m'_{22}$	$-\frac{1}{4}m'_{43}$	$-\frac{1}{2}m'_{44}$	0	$\frac{1}{2}m'_{44}$
b_n		$m'_{14} + \frac{1}{2}m'_{24}$	$\frac{1}{2}m'_{13} + \frac{1}{4}m'_{23}$	$-\frac{1}{4}m'_{42}$	0	$-m'_{41} - \frac{1}{2}m'_{42}$	0
n	7	8	9	10	11	12	
a_n	$\frac{1}{4}m'_{43}$	$\frac{1}{8}m'_{22} + \frac{1}{8}m'_{33}$	$\frac{1}{4}m'_{34}$	$\frac{1}{2}m'_{21} + \frac{1}{4}m'_{22}$	$-\frac{1}{4}m'_{34}$	$\frac{1}{8}m'_{22} - \frac{1}{8}m'_{33}$	
b_n	$-\frac{1}{4}m'_{42}$	$-\frac{1}{8}m'_{23} + \frac{1}{8}m'_{32}$	$-\frac{1}{4}m'_{24}$	$\frac{1}{2}m'_{31} + \frac{1}{4}m'_{32}$	$\frac{1}{4}m'_{24}$	$\frac{1}{8}m'_{23} + \frac{1}{8}m'_{32}$	

The transmission axes of the polarizer and analyzer are assumed to be set at 0 azimuth, parallel to the scattering plane or the plane of incidence.

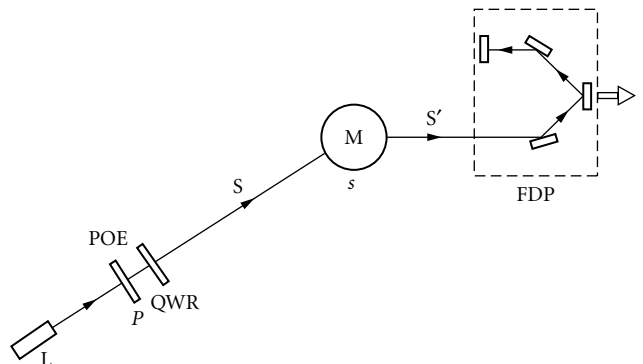


FIGURE 20 Scheme for Mueller-matrix measurement using the four-detector photopolarimeter.¹⁵²

Mueller matrix $\mathbf{M}'$ which differs from $\mathbf{M}$ only by a scale factor. Inasmuch as only the normalized Mueller matrix, with unity first element, is of interest, the unknown scale factor is immaterial. This dual-rotating-retarder Mueller-matrix photopolarimeter has been used to characterize rough surfaces¹⁵⁰ and the retinal nerve-fiber layer.¹⁵¹

Another attractive scheme for Mueller-matrix measurement is shown in Fig. 20. The FDP (or equivalently, any other DOAP) is used as the PSD. Fourier analysis of the output current vector of the FDP, $I(C)$, as a function of the fast-axis azimuth C of the QWR of the input PO readily determines the Mueller matrix $\mathbf{M}$, column by column.^{152,153}

16.9 APPLICATIONS

The applications of ellipsometry are too numerous to try to cover in this chapter. The reader is referred to the books and review articles listed in the bibliography. Suffice it to mention the general areas of application. These include: (1) measurement of the optical properties of materials in the visible, IR, and near-UV spectral ranges. The materials may be in bulk or thin-film form and may be optically isotropic or anisotropic.^{3,22,27–31} (2) Thin-film thickness measurements, especially in the semiconductor industry.^{2,5,24} (3) Controlling the growth of optical multilayer coatings¹⁵⁴ and quantum wells.^{155,156} (4) Characterization of physical and chemical adsorption processes at the vacuum/solid, gas/solid, gas/liquid, liquid/liquid, and liquid/solid interfaces.^{26,157} (5) Study of the oxidation kinetics of semiconductor and metal surfaces in various gaseous or liquid ambients.¹⁵⁸ (6) Electrochemical investigations of the electrode/electrolyte interface.^{18,19,32} (7) Diffusion and ion implantation in solids.¹⁵⁹ (8) Biological and biomedical applications.^{16,20,151,160}

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